

ChE 382: Unit Operations Laboratory

GAS ABSORPTION

IN A PACKED BED

REVISION: Spring 2019v1

Introduction

Gas absorption is a widely used unit operation in chemical engineering. There are numerous process applications where particular components need to be selectively removed from a gas stream. The type of contacting unit for the feed gas and the absorbing liquid can be tailored to the application along with the choice of the absorbent which will have a high selectivity for the desired components along with a high capacity for the absorbed components.

The use of chemically reactive absorption systems has been developed by chemical engineers as a process that shows high selectivity and capacity as compared to absorbents which do not chemically react with the absorbing species. There are numerous applications of this approach in the chemical industry. Examples are the removal of ammonia in the Haber process of ammonia synthesis, separations of acid such as CO_2 and H_2S from natural gas, and CO separation from gas mixtures with H_2 .

There are numerous types of contactors that have been developed to assure a good contact between the gas and liquid streams. This is important to obtain an efficient separation process that requires a minimal size for the absorber. Many of the contactors are similar to those used in distillation. These include bubble cap trays and columns filled with various packings.

Chemical engineers need to be able to design gas absorbers which produce a treated gas of a desired purity with an optimal size and liquid flow. This can be based on existing correlations and, when required, laboratory and/or pilot plant data.

Objective

The objective of this experiment is:

1. To determine fluid flow characteristics in a packed bed gas absorber, including the flooding point.
2. To determine gas absorption characteristics and the overall mass transfer coefficient in the removal of CO_2 from an air stream with water in a packed bed gas absorber.

Theory

There are several good references on packed bed and gas absorption principles and theory: Wankat [1], King [2], McCabe, Smith and Harriott [3], and Perry's Chemical Engineering Handbook [4] and Treybal [5].

Experiments

An Armfield Gas Absorption Column (UOP7) packed with 1-cm Raschig rings will be used for the experiments. A schematic diagram of the gas absorption column is shown in Figure 1. The inside diameter of the column is 3-1/8" (8 cm) and the dimensions of the Raschig rings are OD 1.0 cm, ID 0.8 cm, L 1.0 cm. There is a liquid flow system using a centrifugal pump and a rotameter to determine the liquid flowrate. The system consists of an air supply from the house air and a supply of pure carbon dioxide from a high pressure cylinder. Both gas streams have rotameters to determine the volumetric flowrate. The absorber system is constructed such that the gas flows countercurrent to the liquid.

See Armfield brochure in the Gas Absorption Information folder in Blackboard for more information.

The system has two water manometers to determine the pressure drop across the absorber unit. The process application selected for the study is the absorption of CO₂ from air into water.

The composition of the gas stream is determined volumetrically using a Hempl Apparatus (located on the gas absorption panel). This analytical approach is based on determining the volume of CO₂ absorbed from a gas sample volume, and thus the composition of the gas sample. Details of the Hempl Apparatus are provided in Appendix A.

The liquid-phase concentration of CO₂ is determined by titration. This is a classical analytical technique for determining concentrations of acids and bases in aqueous solution and is described in Appendix B.

Equipment (partial list)

Gas absorption column
50 mL buret and buret clamp
50 mL graduated cylinder
Stopwatch
1-50 mL pipet (if needed)
125 mL erlenmeyer flasks
Thermometer

Chemicals/Materials

Compressed air
Phenolphthalein indicator solution.
0.05 N or 0.01 N sodium hydroxide solution.
Carbon dioxide gas.
Deionized water
1.0 N sodium hydroxide solution

Experimental Procedure

The experimental procedure is split into three separate parts. In part 1, you will characterize the packed bed column and determine flooding for air/water flowrates. In part 2, you will first bring water to equilibrium with an airstream containing CO₂. After equilibrium is reached, you will follow desorption of CO₂ into normal laboratory air.

In part 3, you will study the absorption of CO₂ by water in a column in a continuous operating mode.

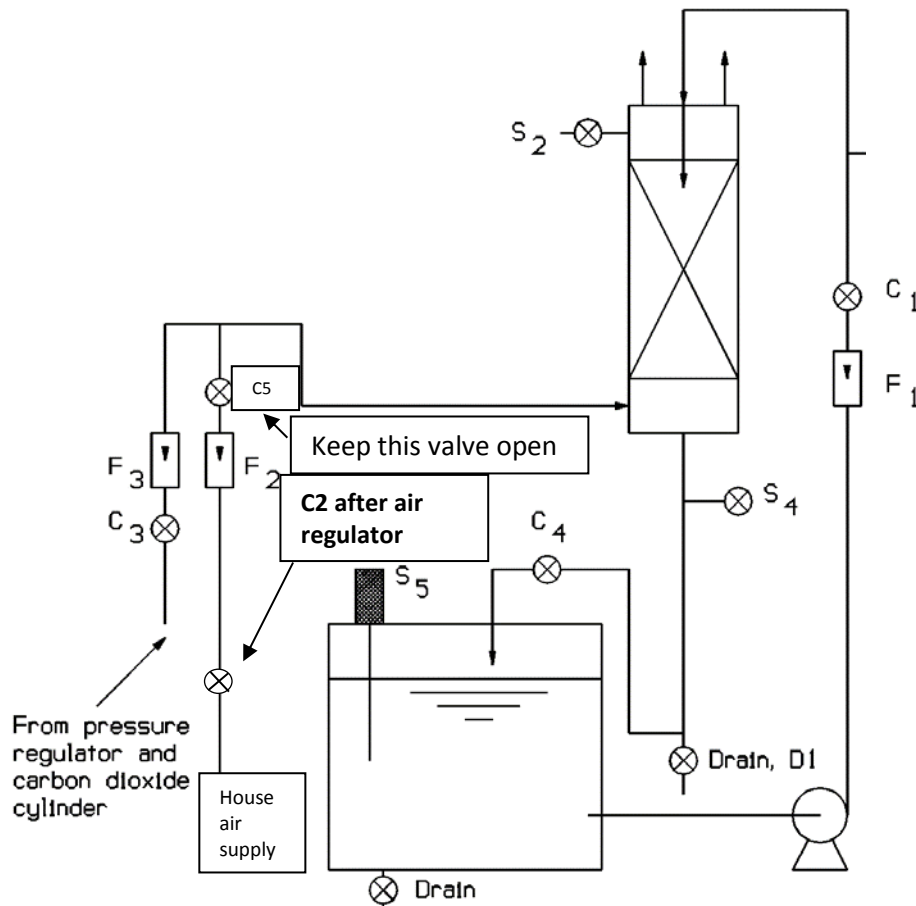


Figure 1. Gas Absorption Column System

Part 1. Characterization of Liquid and Gas Flow in Packed Columns

Packed beds are widely used by chemical engineers to contact gas and liquid phases. Examples are gas absorption and distillation. They are also often used in chemical reactors where the catalyst is impregnated on the surface of the packing. For rational process design, chemical engineers need to be able to characterize the behavior of these contactors. Typically, one wants to be able to describe the pressure drop through the unit to allow sizing of the pumps. Additionally, it may be important to know the heat transfer between the bed and the wall if a highly exothermic or endothermic reaction is occurring. There are numerous packings used for these contactors and numerous calculations to allow estimation of operating parameters such as pressure drop. However, in many cases the correlations do not cover the desired range of operations and it is necessary to generate experimental data.

Packed Bed Flooding

An important decision for the engineer is the selection of relative gas and liquid flowrates in the packed-bed reactor. If the gas flow is much higher than the liquid flow, the gas tends to lift out of the bed and there is poor contact between the phases. Correlations have been developed to guide in process design. An important concept for these design tools is bed flooding. If the pressure drop is measured as a function of gas flowrate for a fixed liquid flowrate, a sharp break in slope will occur. This is defined as the point of flooding and is where the gas flowrate is sufficient to begin to holdup the liquid in the bed. Experience has shown that operating at 50% of flooding gives optimal gas-liquid contacting.

Experimental Procedure

Initially, the void volume of the packed bed should be determined. Two methods are suggested. One method simply measures the amount of water necessary to immerse a given volume of packing. The second method requires determining the density of the solid packing. The void volume can then be obtained from the relationship:

$$\epsilon = \frac{V_{void}}{V_{apparent}}, \text{ or } \epsilon = 1 - \frac{\rho_B}{\rho_s} \quad (1)$$

where ϵ is the void volume defined as volume of voids / total volume of the packed bed, ρ_s is the density of the solid, and ρ_B is the bulk density of the packed particles. Note that equation (1) is sensitive to the value of the porosity; thus care should be used in measuring the ϵ for the various packings.

The void volume of the solid phase includes the voids between the pieces of packing as well as the internal open spaces of each individual piece. It is these spaces through which the gas and liquid must pass. In addition, however, the solid may have internal voids due to inherent porosity. This internal porosity is usually present in very fine pores ($< 1 \mu\text{m}$ in diameter) and depending on the nature of the solid may represent either a negligible or significant fractions of the overall volume of the solid. For metals, glass, or a glazed ceramic, this internal porosity may be entirely absent. When the material has no appreciable fine pore volume, one usually uses a common fluid such as water to measure only the larger void spaces of a solid or the void spaces of a bed of solids; such a displacement measurement is used, for example, to obtain the specific density of a solid. This is done by measuring the amount of water displaced by a bed of solid particles. For this experiment, this may be adequately done using tap water and appropriate graduated cylinders. The density of the packing can then be calculated by dividing the mass by the solids by their displacement volume. The packed bed density can be determined directly by just measuring the volume occupied by a known weight of randomly packed solids.

In the second part of the experiment, the pressure drop across the packed bed is measured as a function of the air and water flow rates. The pressure drop across the packed bed is measured with the manometer connected to the column with appropriate pressure taps.

1. Close the drain valve under the liquid reservoir tank, the control valve, **C4**, and the column drain valve, **D1**. Fill the liquid reservoir tank at the base of the column three-quarters full with deionized water from the tap. Record the temperature of the water.
2. Adjust the air flow rate through flowmeter **F2** to 20 L/min using control valve **C2**. Allow air to flow through the column for 5 min to dry the packing. Measure the pressure drop across the bed for different gas flowrates with no liquid flow.
3. Open control valve **C4**. With gas flow control valves **C2** and **C3** closed, start the liquid pump, open valve **C1** a couple turns for 15-30 s to expel air from the pump, then adjust the water flow through the column to 1.5 L/min on flowmeter **F1** by adjusting flow control valve **C1**.
4. Open gas flow control valve **C2** and adjust flowmeter **F2** to 20 L/min. Again measure the gas pressure drop as you increase the air flowrate until flooding occurs. Repeat the experiment for a several more water flowrates (3 L/min, 6 L/min, 8 L/min).

Part 2a: Bringing water to equilibrium with CO₂ and analysis of CO₂ in water

Close the drain valve under the liquid reservoir tank, the control valve, **C4**, and the column drain valve, **D1**. Fill the liquid reservoir tank at the base of the column three-quarters full with deionized water from the tap. Record the temperature of the water. Place the cover on the tank. Collect a sample and determine its initial CO₂ concentration following the directions in the Analysis Procedure (Appendix B).

Open control valve **C4**. With gas flow control valves **C2** and **C3** closed, start the liquid pump, open valve **C1** a couple turns for 15-30 s to expel air from the pump, then adjust the water flow through the column to 6 L/min on flowmeter **F1** by adjusting flow control valve **C1**.

Adjust the air flow rate through flowmeter **F2** to 20 L/min using control valve **C2**. **NOTE:** Monitor and readjust the air flow rate during the experiment as it has a tendency to drift. Note any readjustments on your data sheets.

Carefully open the main valve on the carbon dioxide cylinder all the way. Slowly adjust the pressure regulating valve on the carbon dioxide regulator to give a gauge reading of approximately 20 psi. Adjust valve **C3** to give a CO₂ flow rate on the flowmeter **F3** corresponding to the flow rate you have been assigned by your faculty instructor, 2 through 5 L/min.

After 10 min of operation, take 50 mL samples at appropriate intervals from **S4** until the dissolved CO₂ values are constant. Analyze the samples following the directions in the Analysis Procedure (Appendix B).

Measure the volume fraction of CO₂ in the air leaving the column (from **S2**) once the dissolved CO₂ levels are constant (see Appendix A).

Part 2b: Desorption of CO₂ into laboratory air

Once the CO₂ values from step 5 are constant, shut off the CO₂ flow at the flowmeter (valve **C3**), decrease the liquid flow rate to 1.5 L/min, open the drain valve, **D1**, under the column, and close the control valve, **C4**. Adjust the drain valve, **D1**, under the column to maintain a liquid level above sample tap, **S4**. Immediately take samples as quickly (e.g. 1 minute intervals) as possible from **S4**. Analyze the samples following the directions in the Analysis Procedure (Appendix B).

Part 3: Absorption of CO₂ by H₂O in a continuous column mode

Close the drain valve under the liquid reservoir tank, the control valve, **C4**, and the column drain valve, **D1**. Fill the liquid reservoir tank at the base of the column full with deionized water. Record the temperature of the water. Collect a sample and determine its CO₂ concentration following the directions in the Analysis Procedure (Appendix B).

Open the control valve, **C4**. With gas flow control valves **C2** and **C3** closed, start the liquid pump. Open valve **C1** a couple turns for 15-30 seconds to expel air from the pump, then adjust the water flow to 1.5 L/min, open the column drain valve, **D1**, and close control valve **C4**. Use the column drain valve, **D1**, to maintain a liquid water height below the air/CO₂ inlet but above the sampling valve **S4**. It is much easier to maintain a liquid level if sample valve **S4** is left open. A bucket can be placed under the sample tubing to collect the water. Adjust the air flow rate through flowmeter **F2** to a value of 20 L/min using control valve **C2**. **NOTE:** Monitor and readjust the air flow rate during the experiment as it has a tendency to drift. Note any readjustments on your data sheets.

Carefully open the main valve on the carbon dioxide cylinder all the way. Slowly adjust the pressure regulating valve on the carbon dioxide regulator to give a gauge reading of approximately 20 psi. Adjust valve **C3** to give a CO₂ flow rate on the flowmeter **F3** corresponding to the flow rate you have been assigned by your faculty instructor, 2 through 6 L/min.

As soon as the CO₂ is turned on, take 50 mL samples from **S4** at 30 second intervals during the first two minutes. Increase the interval between samples to one minute for the next five minutes. The interval can then be increased to five minutes until the concentration reaches a steady state. Analyze the samples following the directions in the Analysis procedure (Appendix B).

Analyze a gas sample from **S2** once steady state is reached (see Appendix A).

Note: Steps 2, 3, and 4 should be done as quickly as possible or the flow rates pre-set, the reservoir again filled with water and the experiment started. This will insure that you have enough water to complete the experiment.

Data Analysis

The analysis section should include:

1. Characterization of gas-liquid absorbers: This should describe the various regimes of gas-liquid contacting typically found in absorbers and how the pressure drop data versus gas flow rate can be used quantitatively to define the various regimes. The discussion should also include the region usually selected for gas absorber operation and the reasons for this selection.
2. Absorption of CO₂ into Water: This is a basic mass-transfer process where the CO₂ is transported across the gas-liquid interface. The theory should propose how the data obtained are used to describe the process in terms of mass-transfer coefficients. The first important step is a mass balance to check the accuracy of the flow rate measurements and the chemical analysis. If the data give a very poor mass balance, the mass-transfer coefficients will have little relation to literature values. The second aspect is to use the data to determine the mass transfer coefficient. There is a mass-transfer coefficient for the transport from the gas phase to the gas-liquid interface and another one for the transport from the interface to the bulk of the liquid. See references for a discussion of these resistances in series. Based on previous studies, you would like to argue that one of the two resistances dominates the transport process. If so, one can write a simple expression:

$$N = ka\Delta C \quad (1)$$

where N is the moles of CO₂ transferred, ka is the mass transfer coefficient times the interfacial area (which we do not know) and ΔC is the driving force (partial pressure, concentration, mole fraction).

SAFETY NOTES

Before starting the experiment, review the Material Safety Data Sheets (MSDS) on NaOH. The sheets can be found in the MSDS notebook located in the laboratory.

Personal protective equipment shall include goggles. Disposable nitrile gloves should be worn when handling NaOH solutions.

Check the safety requirements you need to be aware of when using high pressure gas cylinders.

A small amount of NaOH spilled on the outside of glassware can cause the glassware to be extremely slippery when wet. Be careful.

WASTE DISPOSAL PROCEDURES

All wastes from this experiment can be put into the sewer system without causing any problems. When disposing of wastes into the sewer system, please run additional water from the tap at the same time.

References

- [1] Wankat, P.C., Chapter 13, Chapter 15, "Equilibrium Staged Separations", Prentice-Hall, Inc. 1988.
- [2] King, C.J., Chapter 12, "Separation Processes", McGraw-Hill, 1980.
- [3] McCabe, W. L., Smith, J. C., Marriott, P., Chapter 18, "Unit Operations of Chemical Engineering", 7th Edition, McGraw-Hill, 2005.
- [4] R. H. Perry and D. W. Green, Eds., *Perry's Chemical Engineer's Handbook*, 7th ed. McGraw-Hill, 1997.
- [5] R. Treybal, *Mass Transfer Operations*, 2nd ed. McGraw-Hill, 1980.

Appendix A

Use of Gas Analysis Equipment

Measurement of the Volume Fraction of CO₂ in the Gas Stream

Familiarize yourself with the analysis system for CO₂ in the gas stream; this is described below. The TA or Faculty Instructor will show you how it works. (The absorption globes should already be filled with 1.0 N NaOH.) It is also important for you to understand the operation of the three-way stopcocks; diagrams are given below.

Steps to become familiar with the Hempl Apparatus:

1. Safety: What are the safety precautions that must be taken when working with high pressure gas cylinders?
2. Try out the analysis system by measuring the volume fraction of CO₂ in the ambient air. What do you expect the volume fraction to be? Are the results reasonable?
3. Insure the compressed air line is connected to the unit and the valve on the air line is open (next to the unit). Adjust the air flow rate through the flow meter to 20 L/min using the control valve **C2**. The air flow tends to drift so it is important to monitor it periodically.
4. Reconnect the CO₂ line to the flow meter at the rear of the experiment panel. Adjust the CO₂ flow to 2 L/min.
5. Collect a sample of gas from the top of the column, valve **S2**, and determine the volume fraction of CO₂ in it. Repeat the measurement with a new gas sample.
6. You should continue to measure the volume fraction of CO₂ in gas samples from the top of the column for CO₂ flow rates of 3, 4 & 5 L/min with the air flow set at 20 L/min throughout.

In the following procedure three-way stopcocks are used in sampling the gas stream. The operation of the three-way stopcock is shown in Figure 2.

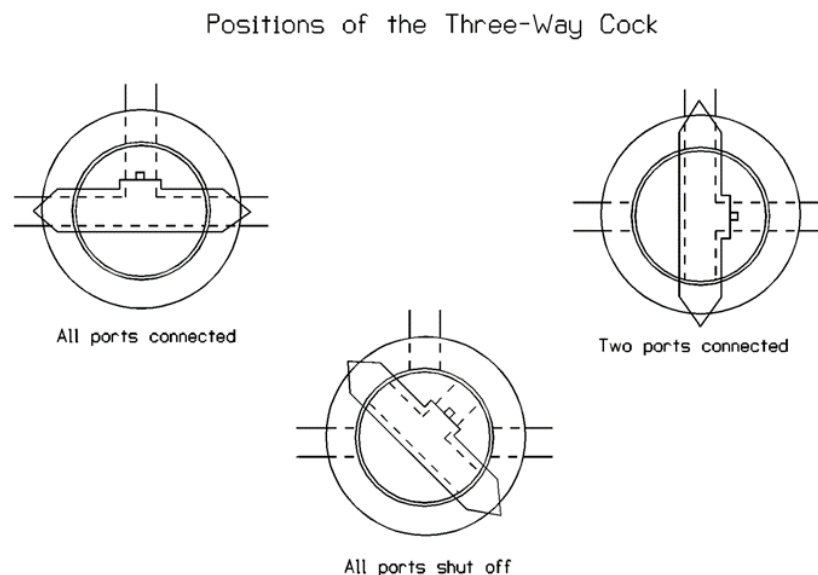


Figure 2. Three-way stopcock positions

Gas Sampling Procedure

1. Fill the two globes of the absorption analysis equipment on the left of the panel with 1.0 N NaOH solution if it has not already been done. Use a small funnel and tubing. NOTE: Wear safety gloves and goggles. Adjust the level in the globes to the "0" mark on the sight tube, by draining liquid through valve Cv into a flask (See A in Figure 3.). This will normally have been done for you. Ask the TA or Instructor to add more NaOH if necessary to bring the level to zero on the measuring tube scale.
2. After 15 minutes or so of steady operation, take samples of gas from sample point S2. Analyze these samples as given in the instructions listed below.
3. Pull the piston of the syringe out to the stop. Take a squirt bottle of deionized water and wet the piston of the syringe. This insures there is a water seal so the gas pulled into the syringe can not leak out to the atmosphere around the syringe piston. This may have to be done several times during the experiment since the water on the piston does evaporate.
4. Flush the sample lines by repeated sucking from the line using the syringe and expelling the contents of the syringe to the atmosphere. Note that the volume of the cylinder is about 100 mL. Estimate the volume of the tubing leading to the syringe. How do you estimate this volume and how many times will you need to purge the line (Steps B & C)?

5. With the absorption globe isolated and the vent to atmosphere closed, fill the syringe from the selected line by drawing the syringe piston out slowly (Step B). Note volume taken into the syringe V1, which should be approximately 40 mL for this particular experiment (See WARNING note below). Wait at least two min to allow the gas to come to the temperature of the syringe.
6. Isolate the syringe from the column and the absorption globe and vent the syringe to atmospheric pressure. Close after about 10 seconds. (Step D).
7. Connect the syringe to the absorption globe. The liquid level should not change. If it does change, briefly open to atmosphere again.
8. Wait until the level in the indicator tube is on zero showing that the pressure in the cylinder is atmospheric.
9. Slowly close the piston to empty the gas in the syringe into the absorption globe. Slowly draw the piston out again (Steps E and F). Why would you want to do this step slowly?

Note the level in the indicator tube.

Repeat steps E and F until no significant change in level occurs. Read the indicator tube marking - V2. This represents the volume of the gas sampled.

WARNING: If the concentration of CO₂ in the gas sampled is greater than 8%, it is possible to suck liquid into the cylinder. This will ruin your experiment and takes time to correct. Under these circumstances, do not pull the piston out to the end of its travel. Stop it at a particular mark, e.g. V1 = 40 on the coarse scale, and read the fine scale.

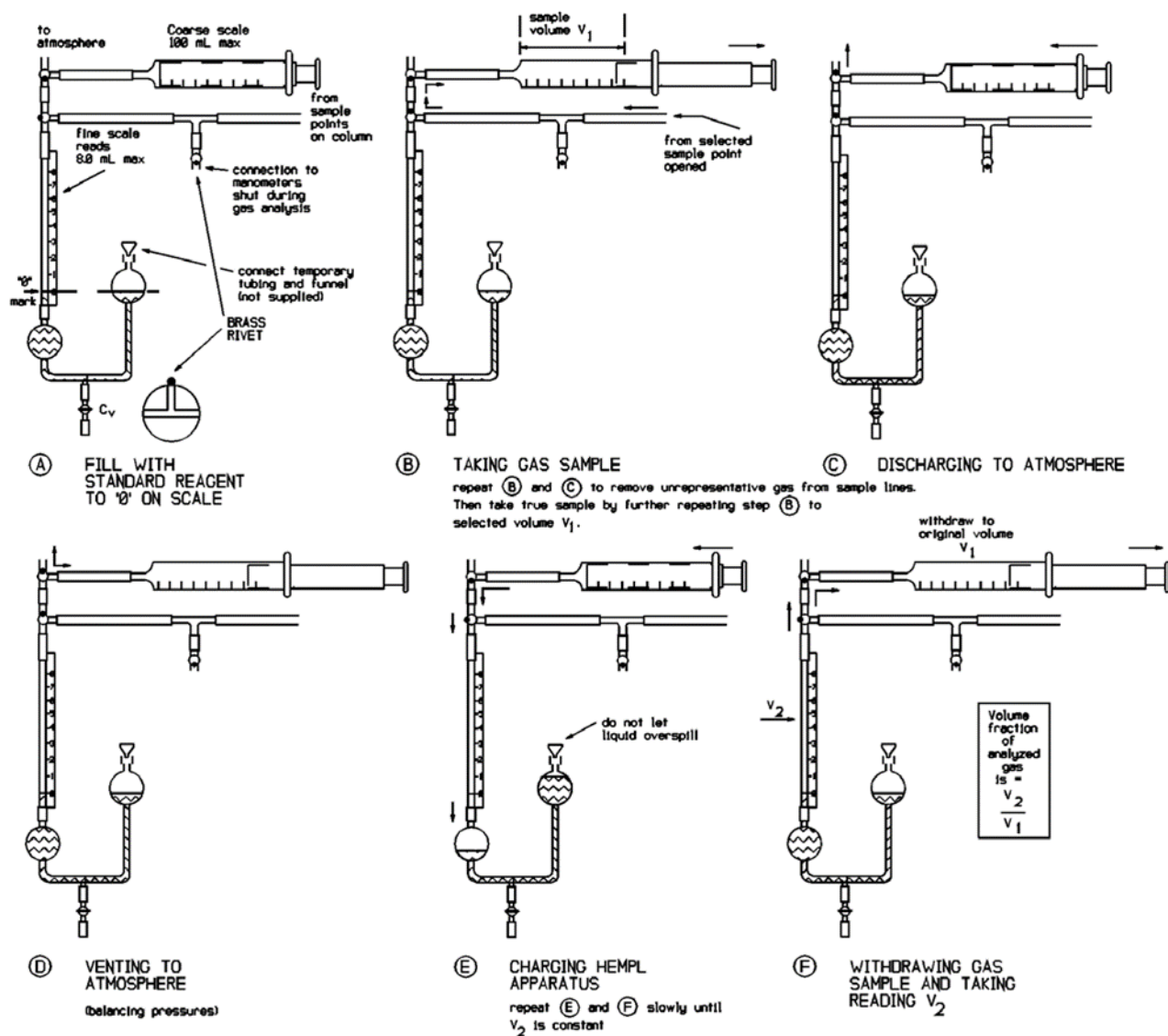


Figure 3. Hempl apparatus for gas analysis

Appendix B

ANALYSIS PROCEDURE FOR CO₂ IN WATER

1. The sodium hydroxide used in the experiment is standard solution of either 0.01 N or 0.05 N NaOH.
2. Collect 50 mL with a graduated cylinder from the liquid outflow point S4. If needed, collect a sample from the reservoir tank. NOTE: Purge the sample line before collecting sample. The sample should be collected with the sample tube below the liquid level in the graduated cylinder.
3. Place the sample in a 125 mL erlenmeyer flask.
4. Add 3-5 drops of phenolphthalein indicator solution. If the sample turns red immediately, no free CO₂ is present. If the sample remains colorless, swirl the flask and titrate with standard NaOH solution. The end point is reached when a definite pink color persists for about 30 seconds - record volume V_B of the NaOH added.
5. The amount of free CO₂ in the water sample is calculated from:

$$C_d \left(\frac{\text{gram mole}}{\text{liter of free CO}_2} \right) = \left(\frac{V_B \times (\text{Conc. Base})}{\text{mL of Sample}} \right)$$

C_{di} = Concentration at inlet (corresponds to conditions at the top of the tower, S6).

C_{do} = Concentration at outlet (corresponds to conditions at the bottom of the tower, S4).

Note: Solubility of CO₂ in water is a strong function of temperature and pH.